Malaria morbidity and pyrethroid resistance after the introduction of insecticide-treated bednets and artemisinin-based combination therapies: a longitudinal study

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Summary

Background Substantial reductions in malaria have been reported in several African countries after distribution of insecticide-treated bednets and the use of artemisinin-based combination therapies (ACTs). Our aim was to assess the effect of these policies on malaria morbidity, mosquito populations, and asymptomatic infections in a west African rural population.

Methods We did a longitudinal study of inhabitants of Dielmo village, Senegal, between January, 2007, and December, 2010. We monitored the inhabitants for fever during this period and we treated malaria attacks with artesunate plus amodiaquine. In July, 2008, we offered longlasting insecticide (deltamethrin)-treated nets (LLINs) to all villagers. We did monthly night collections of mosquitoes during the whole study period, and we assessed asymptomatic carriage from cross-sectional surveys. Our statistical analyses were by negative binomial regression, logistic regression, and binomial or Fisher exact test.

Findings There were 464 clinical malaria attacks attributable to *Plasmodium falciparum* during 17 858 person-months of follow-up. The incidence density of malaria attacks averaged 5·45 (95% CI 4·90–6·05) per 100 person-months between January, 2007, and July, 2008, before the distribution of LLINs. Incidence density decreased to 0·41 (0·29–0·55) between August, 2008, and August, 2010, but increased back to 4·57 (3·54–5·82) between September and December, 2010—ie, 27–30 months after the distribution of LLINs. The rebound of malaria attacks were highest in adults and children aged 10 years or older: 45 (63%) of 71 malaria attacks recorded in 2010 compared with 126 (33%) of 384 in 2007 and 2008 (p<0·0001). 37% of *Anopheles gambiae* mosquitoes were resistant to deltamethrin in 2010, and the prevalence of the Leu1014Phe *kdr* resistance mutation increased from 8% in 2007 to 48% in 2010 (p=0·0009).

Interpretation Increasing pyrethroid resistance of *A gambiae* and increasing susceptibility of older children and adults, probably due to decreasing immunity, caused the rebound and age shift of malaria morbidity. Strategies to address the problem of insecticide resistance and to mitigate its effects must be urgently defined and implemented.

Funding Institut de Recherche pour le Développement and the Pasteur Institute of Dakar.

Lancet Infect Dis 2011; 11: 925–32

Published Online August 18, 2011 DOI:10.1016/S1473-3099(11)70194-3

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Introduction

During the past decade there have been substantial changes in malaria and its control throughout Africa. Depending on the region, mortality due to malaria increased two to six times during the 1990s because of the dissemination of high levels of resistance to chloroquine in *Plasmodium falciparum*.[1–3] However, most African countries only abandoned chloroquine between 2004 and 2008, when large-scale international funding became available for the use of combination therapies.[4,5] During a transitory phase sulfadoxine with pyrimethamine alone or the combination of amodiaquine plus sulfadoxine with pyrimethamine was used in several countries. After 2006, artemisinin-based combination therapies (ACTs) were rapidly deployed,[6] and by 2010 ACTs were the first-line treatment in every malaria-endemic country in Africa.[7] Depending on the region, the change in the first-line treatment of malaria attacks was preceded, followed, or accompanied by the mass distribution of insecticide-treated bednets.[8] Studies in Kenya, Senegal, and The Gambia have shown that these policies substantially reduce malaria morbidity, mortality, and prevalence.[9–12]

Although prompt access to effective drugs prevents most malaria deaths at the community level, even in a context of intense malaria transmission,[13] and the use of insecticide-treated bednets or curtains (ITNs) substantially reduce the burden of malaria,[14] we do not know the extent to which these policies might durably reduce malaria morbidity. Emerging artemisinin resistance of *P falciparum* has been reported in South America and southeast Asia,[15,16] and pyrethroid resistance of *Anopheles gambiae* is increasing in Africa,[17,18] both representing major threats for present malaria control strategies. Evidence of an increase in malaria morbidity in Rwanda, the island of São Tomé, and Zambia in 2009 is of concern because in these three countries scaling up of control measures had resulted in a sharp decline of cases in the previous years.[19,20]

Since 1990, the population of Dielmo, a Senegalese village, has been involved in a long-term study of the relation between the host and the malaria vector.[21] Daily monitoring of fever and monthly mosquito captures have generated a unique dataset, which allows historical analysis of the effect of precisely timed interventions on malaria morbidity and epidemiology. ACTs were introduced for the first-line treatment of uncomplicated malaria attacks in June, 2006, throughout Senegal, by decision of the Senegalese Ministry of Health in accordance with WHO recommendations. Previously, malaria attacks in Dielmo were treated with an oral combination of quinine, quinidine, cinchonine, and cinchonidine (June, 1990, to December, 1994); chloroquine (January, 1995, to October, 2003); or amodiaquine plus sulfadoxine with pyrimethamine (November, 2003, to May, 2006). In this study, we focus on 2007–10, during which ACTs were introduced for the first-line treatment of malaria attacks (June, 2006), and longlasting insecticide (deltamethrin)-treated nets (LLINs; Permanet 2.0) offered to all villagers (July, 2008). Our aim was to assess the effect of these policies on malaria morbidity, asymptomatic parasite carriage, and mosquito populations.

Methods

Participants

Between 1990 and 2010 we did a longitudinal study involving the inhabitants of the village of Dielmo, Senegal, to identify all episodes of fever. Our study included daily medical surveillance with systematic blood testing of individuals with fever and examination of 200 oil-immersion fields on a thick blood film for malaria parasites (about 0·5 µL of blood).

The village is situated in a Sudan-savannah region of central Senegal, on the marshy bank of a small permanent stream, where anopheline mosquitoes breed all year round.[21] Malaria transmission is intense and perennial, with a mean 258 infected bites per person per year during 1990–2006.[22]

Written informed consent was obtained from all participants in our study or the guardians of children younger than 15 years. Our project was initially approved by the Ministry of Health of Senegal and the assembled village population. Approval was then renewed on a yearly basis. Audits were done regularly by the National Ethics Committee of Senegal and ad-hoc committees of the Ministry of Health, the Pasteur Institute (Dakar, Senegal), and the Institut de Recherche pour le Développement (Marseille, France).

Procedures

We gave each individual a unique identification code for our project and prepared a file that contained a photograph, details of family ties, occupation, and precise place of residence on detailed maps of each household with the location of each bedroom. We visited all households daily, and collected nominative information 6 days a week (ie, excluding Sunday) at home on the presence or absence in the village of each individual we had enrolled, their location when absent, and the presence of fever or other symptoms. We systematically recorded body temperature at home three times a week (every second day) in children younger than 5 years, and in older children and adults in cases of suspected fever or fever-related symptoms. In cases of fever or other symptoms, blood testing was done at our dispensary by finger prick, and we provided detailed medical examination and specific treatment. The dispensary created for our project was open 24 h a day, 7 days a week to allow both active and passive case detection.

We treated malaria attacks with combination artesunate plus amodiaquine. We measured treatment efficacy with daily clinical surveillance of patients and with at least one control of parasitaemia between day 7 and day 35 after fever resolved.

During the second week of July, 2008, we offered all villagers LLINs; there were no LLINs in Dielmo before this. We visited households quarterly to confirm ownership of nets and to assess their use and condition. During these visits, we asked all villagers enrolled in our study if they used nets (either LLINs or untreated nets) the night preceding the visit and whether they never, always, or sometimes used nets. We measured vector density and entomological inoculation rate monthly by collecting human-landing mosquitoes, and subsequently testing the circumsporozoite protein rate of collected *Anopheles* spp with ELISA. Each month, we analysed a sample of mosquitoes of the *A gambiae* sensu lato (sl) complex with PCR to establish the species. We measured sensitivity of *A gambiae* sl to pyrethroids in September, 2010, by the standard WHO method[23] and we assessed the presence of the Leu1014Phe *kdr* mutation (*kdr* west) on two samples of 50 *A gambiae* sl in 2007 and 2010 by the method of Martinez-Torres and colleagues.[24]

We assessed efficacy of LLINs by bioassay cones in a random sample of 30 nets (five nets in September, 2010, and 25 nets in April, 2011; one net selected at random per household). We used mosquitoes from cyclic colonies of *A gambiae* maintained at the Institut de Recherche pour le Développement insectarium in Dakar in the cone bioassays in accordance with the WHO pesticide evaluation scheme.[25] For each net, ten batches of five unfed female mosquitoes (2–3 days old) were exposed to 25 cm by 25 cm net samples for 3 min. We measured the knock down and mortality after 3 min and after 24 h.

We assessed the pattern of clinical malaria attacks incidence density, potential exposure to infected mosquitoes, and asymptomatic malaria prevalence between January, 2007, and December, 2010—ie, during the 18 months preceding the introduction of LLINs and the 30 months after this introduction. We monitored the presence of each person in the village daily and calculated incidence rates of clinical malaria attacks as the ratio of the number of clinical malaria attacks

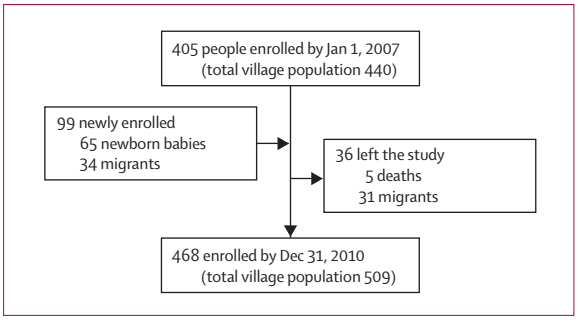


Figure 1: **Study profile**

recorded during a given period divided by the number of followed up person-days under survey during the corresponding period. We derived mean monthly incidence rates by period from the daily incidence rates on the basis of 30·4 days per month. For each *Plasmodium* species, episodes of fever were attributable to malaria when parasite density was higher than an age-dependent threshold calculated with the parasite densities recorded between 2007 and 2010, during our cross-sectional monthly surveys and the fever episodes according to a method described elsewhere.[26] The threshold for *P falciparum* attacks ranged from a maximum of 175 trophozoites per 100 leucocytes in children aged 2 years (about 14 000 parasites per µL of blood) to a minimum of seven in adults aged 60 years or older (about 580 parasites per µL of blood). Blood was taken by finger prick and we examined 200 oil-immersion fields. For clinical attacks from *Plasmodium malariae* we used a cutoff of 35 parasites per 100 leucocytes, and for *Plasmodium ovale* we used a cutoff value of ten, because all infections associated with fever were in children and there was no clear evidence of a better assessment of morbidity attributable to these parasites with age-dependent functions.[27] The malaria prevalence data we assess are those of cross-sectional surveys done at the end of the rainy season in October, 2007, 2008, 2009, and 2010. They include all individuals enrolled in our study that were present in the village at the time of our survey.

Statistical analysis

We did our statistical analyses with negative binomial regression (accounting for the total number of malaria attack cases and the length of the time of follow-up), logistic regression, and binomial or Fisher exact test as appropriate with Stata software (version 11). p less than 0·05 was considered significant.

Role of the funding source

The sponsor of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

We invited all villagers to enrol into our project: 247 at the beginning of our project in June, 1990; 440 in January, 2007; and 509 in December, 2010. Our study cohort was 405 people aged from 60 days to 96 years on Jan 1, 2007, including 301 permanent residents of the village (defined by at least 272 days residence in Dielmo in 2007; figure 1). On Dec 31, 2010, our study cohort was 468 people aged 2 days to 100 years, including 351 permanent residents. 41 villagers refused to participate, including 19 permanent residents.

Between January, 2007, and December, 2010, we followed up 504 villagers during a total of 17 858 person-months (542 889 person-days). We recorded 464 malaria attacks due to *P falciparum* (table), one due to *P ovale*, and three due to *P malariae*. The attacks attributable to *P ovale* and *P malariae* happened before the distribution of LLINs. On average, the incidence density of malaria attacks decreased 5·8 times after the introduction of LLINs (figure 2). However, *P falciparum* malaria community level incidence density increased in the last few months of 2010: 4·57 (95% CI

	0–4 years	5–9 years	10–14 years	15–29 years	30–44 years	≥45 years	Total
January, 2007, to July, 2008							
Follow-up days	38 124	34 136	29 769	41 290	23 635	32 763	199 717
Malaria attacks	112	130	54	33	21	8	358
Monthly incidence	0·089	0·116	0·055	0·024	0·027	0·007	0·055
August, 2008, to August, 2010							
Follow-up days	53 328	50 823	41 704	69 632	33 822	49 970	299 279
Malaria attacks	3	15	10	8	3	1	40
Monthly incidence	0·002	0·009	0·007	0·003	0·003	0·001	0·004
September, 2010, to December, 2010							
Follow-up days	7458	6576	5908	9647	6108	8196	43 893
Malaria attacks	13	13	13	14	9	4	66
Monthly incidence	0·053	0·060	0·067	0·044	0·045	0·015	0·046

Table: **Number of follow-up days and *Plasmodium falciparum* morbidity by age group and control period**

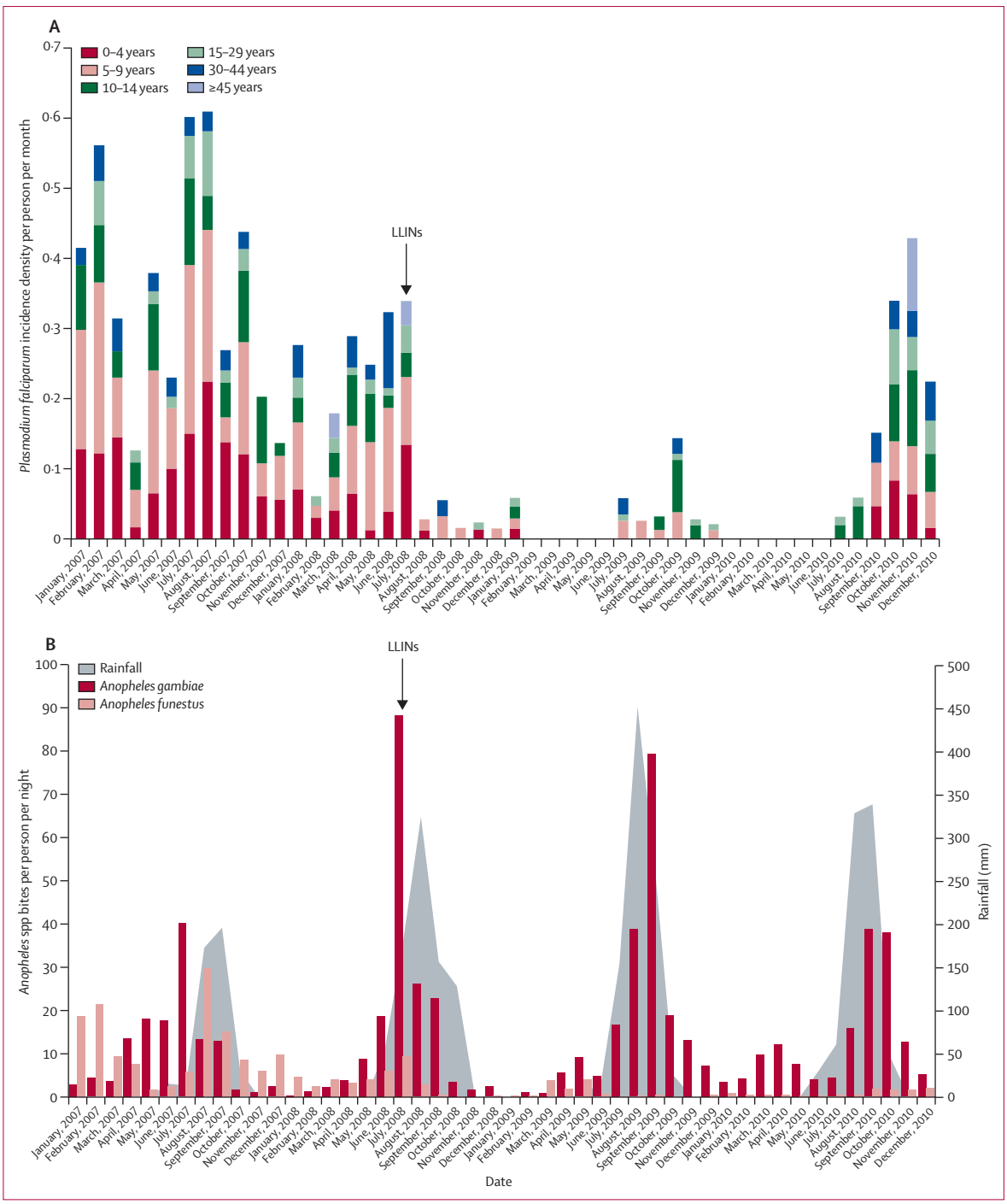


Figure 2: **Monthly incidence density of *Plasmodium falciparum* malaria attacks (A) and monthly variations of the mean daily *Anopheles gambiae* sensu lato and *Anopheles funestus* human biting rate (B)**
LLINs=introduction of longlasting insecticide-treated net.

3·54–5·82) per 100 person-months between September, 2010, and December, 2010, compared with 0·41 (0·29–0·55) between August, 2008, and August, 2010 (p=0·0001 by two-sided binomial exact test), and 5·45 (4·90–6·05) between January, 2007, and July, 2008, before the introduction of LLINs (p=0·19 by two-sided binomial exact test). The rebound of malaria attacks was most notable in children aged 10–14 years and adults—we recorded 45 (63%) of 71 attacks in these groups in 2010 but only 126 (33%) of 384 in 2007 and

2008 (p<0·0001 by Fisher exact test). When we controlled for the effect of age and seasonal variations in a negative binomial regression analysis, the incidence rate ratio of *P falciparum* clinical malaria attacks from August, 2008, to August, 2010, compared with January, 2007, to July, 2008, was 0·07 (95% CI 0·05–0·10; p<0·0001) and from September to December, 2010, compared with January, 2007, to July, 2008, was 0·84 (0·59–1·21; p=0·352; webappendix pp 1–3). When we accounted for all cases of fever associated with *P falciparum*, whatever the level of parasitaemia, we noted similar patterns: the incidence rates of fever cases associated with malaria parasites in older children and adults during the most recent period returned to levels close to those noted before the introduction of LLINs (webappendix pp 4–8). We did not record therapeutic failure with ACT during the whole study.

In 484 person-nights of captures over 48 months, we caught 6910 *A gambiae* sl, 1652 *Anopheles funestus*, 101 *Anopheles pharoensis*, 99 *Anopheles ziemanni*, and seven *Anopheles wellcomei*. The mean circumsporozoite protein rates from January, 2007, to July, 2008, were 3·0% for *A gambiae* sl and 3·0% for *A funestus*; from August, 2008, to August, 2010, were 0·9% (odds ratio [OR] 0·29, 95% CI 0·18–0·45; p<0·0001; with January, 2007, to July, 2008, as reference) for *A gambiae* sl and 0·4% (OR 0·14, 0·02–1·04; p=0·054) for *A funestus*; and from September to December, 2010, were 1·9% (OR 0·60, 0·32–1·13; p=0·11) for *A gambiae* sl and 1·1% (OR 0·36, 0·05–2·80; p=0·33) for *A funestus*. *A funestus* decreased substantially after the introduction of LLINs but the human biting rate of *A gambiae* sl remained high. The entomological inoculation rate we calculated from the monthly values of the human biting rate and circumsporozoite protein rates of *A gambiae* sl and *A funestus* was 232 infected bites per person per year in 2007, 155 in 2008, 50 in 2009, and 89 in 2010 (58 from September to December, 2010). Each year, *A gambiae* sensu stricto (ss) represented about 80% of *A gambiae* sl during the rainy season, and only about 15% during the dry season when *A arabiensis* was the main species we identified. The parity rate (proportion of females that had laid eggs) of *A gambiae* sl decreased substantially during the first 2 years after the distribution of LLINs, but returned to initial levels in 2010 (figure 3). Malaria vectors were more aggressive between 2100 h and 2300 h after the introduction of LLINs, although most bites were still after midnight (figure 4).

The frequency of the *kdr* west allele increased from 8% in 2007 to 48% in 2010 (p=0·0009 by Fisher exact test). Sensitivity tests to insecticides on *A gambiae* sl mosquitoes we collected in September, 2010, showed 63% sensitivity to deltamethrin, 60% to lamdacyhalothrine, 43% to permethrin, 61% to DDT, 100% to fenithrotion, and 100% to bendiocarb. A random sample of 30 LLINs from villagers confirmed their longlasting insecticide activity (84% average efficacy, only five bednets with efficacy of less than 80%).

Ownership of bednets (webappendix pp 9–10) in our study population after the introduction of LLINs was 98% in 2008 (LLINs 93%, untreated nets 5%), 83% in 2009 (LLINs 79%, untreated nets 4%), and 79% in 2010 (LLINs 75%, untreated nets 3%). Bednets were used regularly (ie, during the previous night and reported always using bednets; webappendix pp 11–12) by 79% of our study population in 2008 (LLINs 76%, untreated nets 4%), 60% in 2009 (LLINs 58%, untreated nets 2%), and 61% in 2010 (LLINs 58%, untreated nets 3%). In September, 2010, the proportion of nets in good state (ie, no holes or only one hole) was 93% (282 of 303), and LLINs were used every night by 70% of adults aged 30 years or older, 63% of adults aged 15–29 years, 72% of children

See Online for webappendix

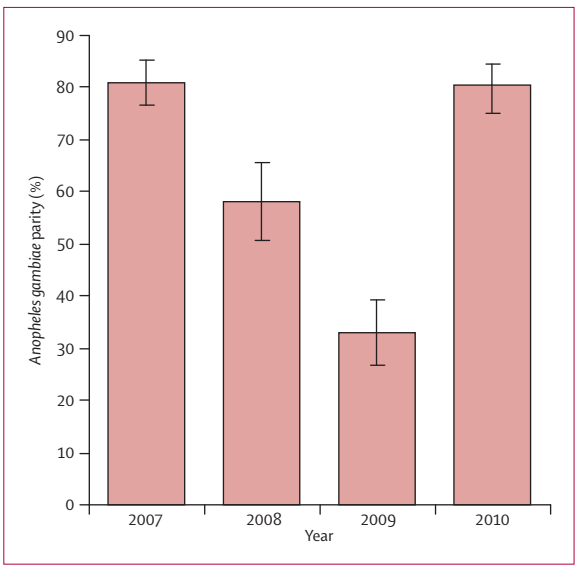


Figure 3: **Annual variation of the parity rate of *Anopheles gambiae* sensu lato**

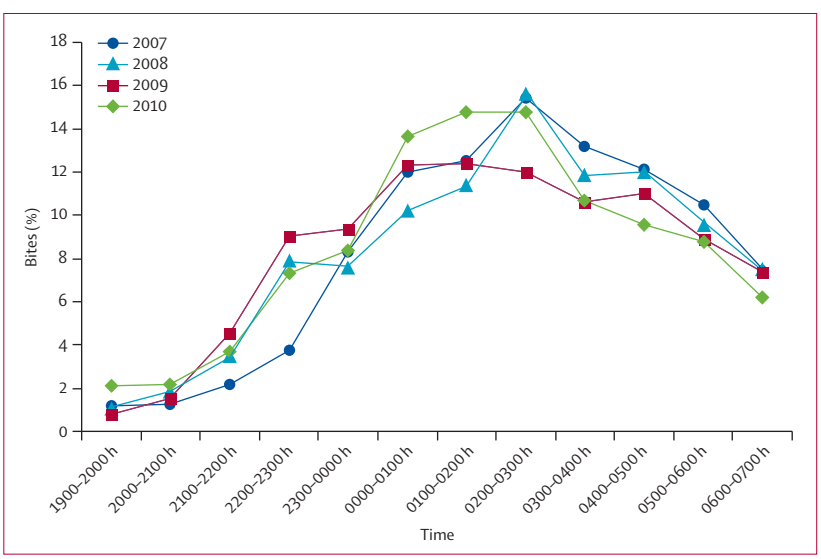


Figure 4: **Biting cycle of malaria vectors before and after the introduction of LLINs**
Cumulated number of bites of *Anopheles gambiae* sensu lato and *Anopheles funestus* per h by total number of bites per night×100. LLIN=longlasting insecticide-treated net.

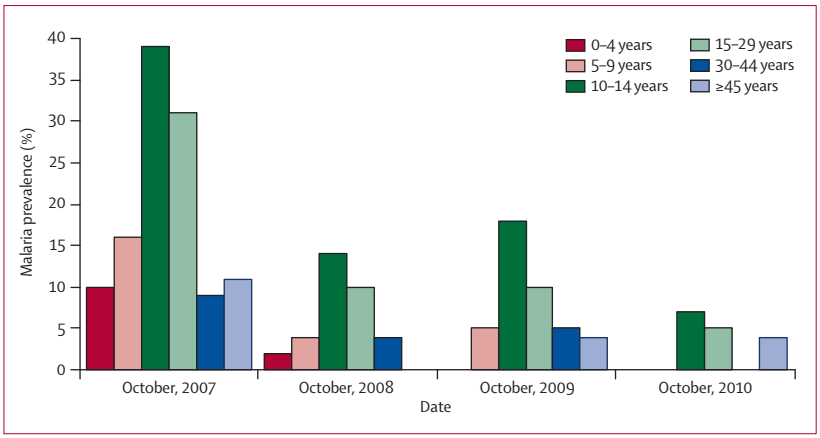


Figure 5: **Prevalence of *Plasmodium falciparum* malaria by age group during our October cross-sectional surveys**
33–81 people were tested at each survey for each age group.

aged 10–14 years, 72% of children aged 5–9 years, and 79% of children aged 0–4 years.

Mean malaria prevalence (figure 5) was 16·3% in 2007, 4·8% in 2008, 5·1% in 2009, and 2·7% in 2010. 99% of infections were due to *P falciparum* and 1% due to *P malariae*. The gametocyte rate of *P falciparum* was 3·2% in 2007, 1·2% in 2008, 1·5% in 2009, and 1·3% in 2010.

Discussion

The incidence density of malaria attacks in Dielmo decreased substantially within 3 weeks of the distribution of LLINs in July, 2008. From August, 2008, to August, 2010, *P falciparum* malaria morbidity was 13-times lower than in the period from January, 2007, to July, 2008, when ACT alone was used for malaria control. We recorded a similar decrease in malaria morbidity when including in our analysis the period from June to December, 2006—ie, the 7 months immediately after the introduction of ACT (data not shown). However, from September to December, 2010 (ie, 27–30 months after the introduction of LLINs), the incidence density of malaria attacks returned to high levels, particularly in adults and older children in whom malaria morbidity was even higher than during the period that preceded the introduction of LLINs.

The effect of the introduction of LLINs on the survival of *A gambiae* ss and *Anopheles arabiensis* in 2008 and 2009 and the change in 2010 when the parity of these vectors returned to preintervention levels suggest that pyrethroid resistance could be a major factor in the increase in malaria morbidity in 2010. The use of pyrethroid insecticides for malaria vector control has increased in the past decade through the scale-up of LLIN distribution programmes and indoor residual-spraying campaigns, and expectedly this has led to a very rapid spread of pyrethroid resistance in the major malaria vectors.[17] In Dielmo, the proportion of *A gambiae* sl mosquitoes with the *kdr* west resistance allele increased from 8% in 2007 to 48% in 2010, which is consistent with unpublished data from the National Malaria Control Programme recording very low levels of resistance to pyrethroids in most parts of Senegal in 2006–07 but high levels in 2010 (Konate L, Dakar University, personal communication). This clearly shows the strong selective pressure exerted by the 6 million LLINs that have been distributed in Senegal during this period.[11,28]

Pyrethroid resistance can account for reduced efficiency of LLINs across all age groups, but does not explain the increased incidence density of malaria attacks in older children and adults to similar or even higher levels than before the deployment of LLINs. The entomological inoculation rate increased in 2010 compared with 2009, but it remained much lower than before the deployment of LLINs. The increase in malaria morbidity in adults and older children is notable and suggests either a recent increase in exposure to malaria vectors in older age groups or a decrease in protective immunity. The former hypothesis is only marginally supported by our data. The use of LLINs was high in all age-groups, and we recorded only mild changes in the use of bednets in Dielmo in 2010 compared with 2009, thereby excluding behavioural bias.[29] Furthermore, although the proportion of mosquitoes biting between 2100 h and 2300 h increased after the introduction of LLINs, the actual number and percentage of the bites during the first part of the night remained low.

There are strong arguments in favour of the postulated decrease in protective immunity to explain the shift in age and increase in incidence of attacks in 2010. It is generally agreed that the persistence of clinical immunity acquired during early childhood is dependent on sustained exposure and that immunity decreases when exposure to malaria is discontinued.[30–32] Studies in Kenya showed that the mean age of people with clinical attacks increased steadily as exposure declined.[32] The extent to which the deployment of ITNs might lead to a shift in child morbidity and potential mortality has been the matter of much debate in the late 1990s and early 2000s.[14,33–35] In 2011, there are still few data on the long-term effect of ITNs in regions of intense malaria transmission. Two studies[36,37] in regions of high seasonal transmission in Burkina Faso and Ghana did not identify evidence of a shift in child mortality from younger to older children. However, all controlled trials were either short term (1–2 years), or enrolled only young children, thereby preventing the measurement of a possible rebound effect or a shift toward older age groups in the susceptibility to clinical malaria. Furthermore, these studies were done before the introduction of ACTs, when chloroquine was still the first-line treatment for malaria despite high levels of resistance. A Cochrane review[14] suggests that ITNs reduced the incidence of malaria attacks by 50% compared with no nets, and 39% compared with untreated nets—ie, a much lower reduction than that we recorded in Dielmo during the first 2 years of our follow-up, and reduced malaria

Panel: Research in context

Systematic review

We searched PubMed with the terms "malaria", "morbidity", "Africa", "artem", and "LLIN". We did not identify any cohort studies measuring the incidence density of clinical malaria attacks at the community level before and after the introduction of longlasting insecticide-treated nets (LLINs) in an African population receiving artemisinin-based combination therapy (ACT), or studies measuring malaria morbidity before and after the emergence of *Anopheles gambiae* resistance to pyrethroids in an African population using LLINs.

Interpretation

Daily monitoring during 4 years of 504 inhabitants of a Senegalese village and monthly mosquito captures generated a unique dataset, which allowed an integrated analysis of the effect of precisely timed interventions on malaria morbidity and epidemiology. Deploying LLINs in this community receiving prompt treatment with ACT for parasitologically confirmed clinical malaria attacks immediately resulted in a decrease of malaria morbidity and asymptomatic parasite carriage in all age groups. However, 2 years after the deployment of LLINs, we recorded a rebound of malaria morbidity temporally related to the emergence of pyrethroid resistance in *Anopheles gambiae*. In older children and adults, malaria morbidity became even higher than before the deployment of LLINs, suggesting a decrease in protective immunity. Since most infections had become symptomatic in all age groups and ACT had remained fully effective, parasite rates in this community continued to decline to a state that could be thought approaching pre-elimination. However, even very low gametocyte rates in this now poorly immune community seemed sufficient to maintain a high entomological inoculation rate because of the unique vectorial capacity of *A gambiae*. These findings are of great concern, since they support the idea that insecticide resistance might not permit a substantial decrease in malaria morbidity in many parts of Africa where *A gambiae* is the major vector and acquired clinical immunity is a key epidemiological factor.

prevalence by only 13%, again much lower than in Dielmo. These findings support the idea that the efficacy of ITNs might vary with the efficacy of drugs used for first-line treatment of malaria attacks. Consequently, the medium-term and long-term indirect effects of ITNs on clinical immunity might be more substantial with the deployment of ACTs than when chloroquine was used despite high levels of resistance and thus less effect on parasite prevalence and morbidity. From June, 2006, to July, 2008, with ACT as the only control measure, malaria morbidity in the young children of Dielmo was four-times lower than during the first years of our project when an oral combination of quinine, quinidine, cinchonine, and cinchonidine, and then chloroquine were used for treating malaria attacks.[33]

Therefore, the most probable scenario in Dielmo is that the very potent effect of ITNs, combined with effective clearance of parasites by ACTs, has led to a major decrease in incidence of malaria attacks over a period of 2 years. Thereafter resistance to pyrethroids augmented exposure to *A gambiae* sl, which, in a community whose immunity had meanwhile declined, led to a rebound in malaria morbidity because almost all infections had become symptomatic in all age groups. ACTs had remained fully effective, and parasite rates in the community declined to a state that could be thought approaching pre-elimination. However, even very low gametocyte rates in this now poorly immune community seemed sufficient to infect a substantial proportion of (mostly pyrethroid-resistant) *A gambiae* mosquitoes. This maintained a high entomological inoculation rate due to the unique vectorial capacity of this African malaria vector.[38,39]

Our findings, in a closely observed population, show the high efficacy and substantial affect of combining effective vector control and effective case management, and support the reduction of the burden of malaria recently recorded in Senegal and elsewhere in Africa (panel).[9–12,19,20,40] Unfortunately, our findings also show the threat posed by insecticide resistance to the sustained effect of this approach, and the speed at which changes can happen. These findings are of great concern, since they support the idea that present methods and policies will not sustain, at least in older children and adults, a substantial decrease in malaria morbidity in many parts of Africa where *A gambiae* is the major vector and acquired clinical immunity is a key epidemiological factor. Strategies to address the problem of insecticide resistance and to mitigate its effects must be urgently defined and implemented.

Contributors

J-FT designed the study and supervised data analysis. AT, CS, and J-FT supervised field data collection. ON and CB did the entomological studies. ND, ABL, AB, and FDS did the clinical studies. FD-B did the parasitological studies. ND, JF, CR, CM, and CR analysed clinical, entomological, and parasitological data. AT-B, DR, PD, OM-P, and CR contributed to the design and monitoring of the study. J-FT and ND wrote the paper with the contribution of PD, OM-P, and CR.

Conflicts of interest

We declare that we have no conflicts of interest.

Acknowledgments

We thank the villagers of Dielmo for their continuous support of our project. We thank Sian Clarke for very useful comments on an earlier draft of our paper.

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